

L3 Internship on Elliptic Curves - Erwan Rousseau

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1 Motivation

Elliptic curves have been used in a wide range of scientific disciplines since XIXth century when the term was first used. However, elliptic curves are defined by cubic polynomial equations which have been studied since Antiquity, by Ancient Greek mathematicians such as Euclide and Apolonius of Perga and later by Medieval Arab mathematicians such as Al-Hassar and Khayyam. Elliptic curves theory emerged from the work of numerous notorious mathematicians such as Fermat, Euler, Lagrange, Legendre, Abel, Jacobi, Weil and Wiles to name but a few.

Elliptic curves were historically introduced within the context of algebraic geometry and number theory. Yet, they also revealed their potential in other

scientific fields such as physics through the lens of mechanics or in finance where they model complex stochastic phenomenon which cannot be captured by linear or normal models. Besides, elliptic curves play a pivotal role in computer science, especially in error-correcting codes and in cryptography. Indeed, the question of rational points arises in cryptography when solving polynomial equations. Hence, elliptic curves offer a security level equivalent to RSA but with shorter keys thanks to Elliptic Curve Cryptography (ECC algorithm). Similarly, using elliptic curve leads to error-correcting algorithms more efficient and more secure.

2 Algebraic varieties

We give here some definitions inspired by Silverman [1].

Let us fix a natural integer n . We note A_n the ring $\mathbf{Q}[X_1, \dots, X_n]$. Observe that we place ourselves in the field $\mathbf{Q}$ in order to make the proof of the Mordell-Weil Theorem easier, but many things could be generalised to any number field. Moreover, we write $\bar{\mathbf{Q}}$ the algebraic closure of $\mathbf{Q}$.

2.1 Affine varieties

The n -dimensional affine space is $\bar{\mathbf{Q}}^n$. This is an affine variety.

Definition 2.1.1. For any subset S of A_n , we pose the affine subvariety:

$$V(S) = \{(x_1, \dots, x_n) \in \bar{\mathbf{Q}}^n, \forall P \in S, P(x_1, \dots, x_n) = 0\}.$$

In particular, $V(S) = V(\langle S \rangle)$, where $\langle S \rangle$ is the ideal generated by S .

A point x is rational on $V(S)$ over $\mathbf{Q}$ if x belongs to $V(S) \cap \mathbf{Q}^n$. When there is no ambiguity, we only say that x is rational.

Hence, one can see an affine variety as a collection of polynomial equations.

Conversely, we define some ideals thanks to subvarieties of $\bar{\mathbf{Q}}^n$:

Definition 2.1.2. For any affine subset V of $\bar{\mathbf{Q}}^n$, we pose the ideal:

$$I(V) = \{P \in A_n, \forall (x_1, \dots, x_n) \in V, P(x_1, \dots, x_n) = 0\}.$$

Definition 2.1.3. The Zariski topology over $\bar{\mathbf{Q}}^n$ is defined by taking the affine subvarieties as closed subsets.

2.2 Projective varieties

We will be interested in projective varieties, so we give projective analogous of the previous definitions.

First of all, let us denote $\mathbf{P}^n$ the projective n -space, i.e. the space $\bar{\mathbf{Q}}^{n+1} \setminus \{0\}$ quotiented by the colinearity equivalence relation. Similarly, we write $\mathbf{P}^n(\mathbf{Q})$ the quotient of $\mathbf{Q}^{n+1} \setminus \{0\}$ by the colinearity relation.

We cannot consider an arbitrary polynomial P from A_{n+1} when dealing with a projective space because $P(x_0, \dots, x_n)$ is not well defined for any point $(x_0, \dots, x_n)$. This is why we need homogeneous polynomials.

Definition 2.2.1. A polynomial P from A_{n+1} is said to be homogeneous of degree d if:

$$\forall \lambda \in \mathbf{Q}, P(\lambda X_0, \dots, \lambda X_n) = \lambda^d P(X_0, \dots, X_n).$$

We note A_n^0 the set of homogeneous polynomials from A_{n+1} .

We can now adapt previous definitions:

Definition 2.2.2. For any subset S of A_n^0 , we pose the projective subvariety:

$$V(S) = \{(x_0, \dots, x_n) \in \mathbf{P}^n, \forall P \in S, P(x_0, \dots, x_n) = 0\},$$

Definition 2.2.3. For any projective subvariety V of $\mathbf{P}^n$, we pose the homogeneous ideal I , i.e. generated by homogeneous polynomials:

$$I(V) = \langle \{P \in A_n^0, \forall (x_0, \dots, x_n) \in V, P(x_0, \dots, x_n) = 0\} \rangle.$$

Proposition 2.2.4. *The projective space $\mathbf{P}^n$ can be viewed as the affine space $\bar{\mathbf{Q}}^n$ with an added hyperplane at infinity.*

Proof. Let us write U the subspace of $\mathbf{P}^n$ of equation $x_0 \neq 0$. By mapping $(x_0, \dots, x_n)$ to $(x_1/x_0, \dots, x_n/x_0)$, U is isomorphic with the affine space $\bar{\mathbf{Q}}^n$. Then, the complementary of U is the projective hyperplane of equation $x_0 = 0$. $\square$

3 Elliptic curves

3.1 Abstract group law

Definition 3.1.1. A plane affine (resp. projective) curve is an affine (resp. projective) subvariety in $\bar{\mathbf{Q}}^2$ (resp. $\mathbf{P}^2$) defined by one polynomial.

In particular, we will be interested in cubic curves, i.e., curves defined as $V(P)$ where P is a homogeneous polynomial of A_2^0 of degree 3. Therefore, let us fix such a cubic $\mathcal{C}$ defined by a homogeneous polynomial.

Remark. Observe that restricting such a polynomial P to the affine space $\bar{\mathbf{Q}}^2$ gives us a polynomial (not necessarily homogeneous) of two variables (by imposing $Z = 1$ for instance). Conversely, we can homogenize a polynomial P of A_2 by multiplying each monic by Z at the power of the degree of P less the degree of the monic. Hence, we can consider both a polynomial from A_2 such as $Y^2 - X^3 - X - 1$ or its homogenized polynomial $Y^2Z - X^3 - XZ^2 - Z^3$ from A_2^0 depending on whether we restrict ourselves to the affine plane $\bar{\mathbf{Q}}^2$ or to the whole projective space $\mathbf{P}^2$.

Definition 3.1.2. A point $a \in \mathcal{C}$ is said to be singular if:

$$\partial_1 P(a) = \partial_2 P(a) = \partial_3 P(a) = 0,$$

where P is the homogeneous polynomial defining V .

If there is no singular point, $\mathcal{C}$ is smooth.

From now on, we suppose $\mathcal{C}$ to be smooth. This gives us the following useful result:

Theorem 3.1.3. *By applying linear transformation to $\bar{\mathbf{Q}}^2$, the equation of $\mathcal{C}$ can be taken in the canonical affine form $y^2 = x^3 + \alpha x + \beta$ where α and β belong to $\mathbf{Q}$.*

Remark. The canonical form is also available in other fields such as number fields). However, when considering fields of characteristic 2 or 3, some equation cannot be transformed into this canonical form. Fortunately, there exists another canonical form, more complex, which can be used in any field.

From now on, we fix α and β as in Theorem 3.1.3.

We aim at endowing $\mathcal{C}(\mathbf{Q})$ (the set of rational points on $\mathcal{C}$) with a group law (see Cassels [2, Chapter 7]). Let us fix $o = (0, 1, 0)$, a rational point on $\mathcal{C}$ which is going to play the role of the neutral element (fixing such a point makes $\mathcal{C}$ to be an elliptic curve).

Observe that o is a point at "infinity". Indeed, if we decompose $\mathbf{P}^2$ as the union of $\bar{\mathbf{Q}}^2$ (imposing $z = 1$) and a projective hyperplane of equation $z = 0$, $(0, 1, 0)$ is the only point of $\mathcal{C}$ in the projective space. It is therefore convenient to see $\mathcal{C}$ as a union of an affine curve and a point at infinity.

In order to build a law $+$, we need to deal with intersection between $\mathcal{C}$ and lines.

Proposition 3.1.4. *If a and b are two rational points then the line (a, b) (which is considered as the tangent of $\mathcal{C}$ in a when $a = b$) meets $\mathcal{C}$ in a third rational point.*

Proof. Let a and b be two rational points and D be a polynomial such that $(a, b) = V(D)$.

First of all, let us suppose a and b different from o . Then, they belong to the affine space $\bar{\mathbf{Q}}^2$.

Let us write $y = ux + v$ the equation of D (which is supposed non vertical for now). Then, the intersection of D and $\mathcal{C}$ is given by the roots of the equation:

$$u^2x^2 + 2uvx + v^2 = x^3 + \alpha x + \beta.$$

Since it is a polynomial equation of degree three with two rational roots (given by a and b), it has a third rational root by Galois theory.

Now, suppose D vertical. Then, its affine equation is written $x = k$, where k belongs to $\mathbf{Q}$. The intersection of D and $\mathcal{C}$ is given by the roots of the equation:

$$y^2 = k^3 + \alpha k + \beta,$$

in the affine space.

So there are two roots in the affine space. The homogenized equations of D and $\mathcal{C}$ are respectively $x = kz$ and $y^2z = x^3 + \alpha xz^2 + \beta z^3$. Then, in the hyperplane at infinity, we get $x = 0$ and $x^3 = 0$. Therefore, $\mathcal{C}$ meets D at o .

If $a = o$ and $b \neq o$, then the equation of D is of the form $x = k$ where k is a rational constant and we finish as above.

Now, suppose $a = b = o$. We need to change our point of view by considering a new affine space: the affine space of equation $y = 1$ (which is a neighborhood of o). Then the equation of $\mathcal{C}$ becomes $P(x, z) = x^3 + \alpha xz^2 + \beta z^3 - z = 0$. Since $\nabla P(0, 0) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$, the equation of the tangent of $\mathcal{C}$ in $(0, 0)$ is $z = 0$. Therefore, the intersection of $\mathcal{C}$ and D is given by the equation $x^3 = 0$. Thus, o is the only point of the intersection, of multiplicity 3.

□

Definition 3.1.5. Let a and b be two rational points. The line (a, b) cuts $\mathcal{C}$ in a , b and a third rational point c . Similarly, the line (o, c) cuts $\mathcal{C}$ in a third rational point $a + b$ (see Figure 1). When $a = b$, (a, b) is the tangent of $\mathcal{C}$ in a .

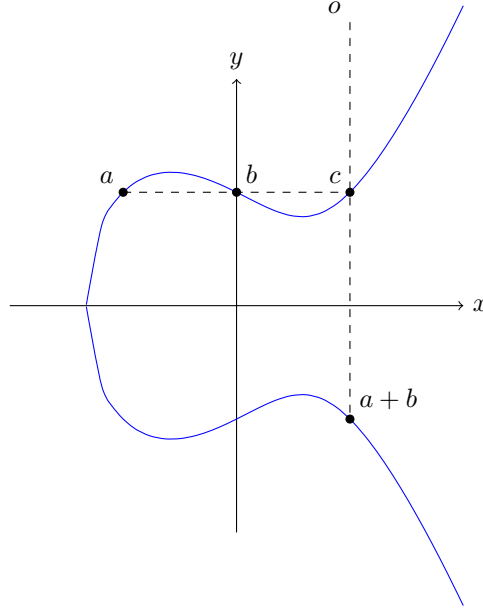


Figure 1: Group law over $y^2 = x^3 - x + 1$

Proposition 3.1.6. $(\mathcal{C}(\mathbf{Q}), +)$ is an abelian group (called the Mordell group).

Proof. First of all, it is straightforward that $+$ is commutative and that o is neutral.

Let a , b and c be rational points.

Let us build $-a$. We define $-a$ as the third intersection of (a, o) with $\mathcal{C}$.

It remains to show that $+$ is associative.

Let us write f_1 the linear form (homogeneous polynomial of degree 1) which is zero on a and b ($f_1(x) = 0$ is the equation of the line (a, b)). Similarly, we

write f_2 the linear form which is zero on o and d , the third point of $\mathcal{C}(\mathbf{Q}) \cap (a, b)$ (where f_1 is zero). We pose:

$$g : \begin{array}{ccc} \mathcal{C} & \longrightarrow & \mathbf{P}^1 \\ a' & \longmapsto & (f_1(a'), f_2(a')) \end{array} .$$

Observe that g is well defined since f_1 and f_2 are homogeneous of degree 1. Moreover, g can be seen as the rational function f_1/f_2 . Hence, a, b can be seen as zeros of g while o and $a + b$ are poles. However, d is neither a zero nor a pole of g since zeros of f_1 and f_2 cancel out. Indeed, since $\mathcal{C}$ is smooth, we can use the implicit function theorem to get a function $\gamma : \bar{\mathbf{Q}} \rightarrow \bar{\mathbf{Q}}$ such that the equation of $\mathcal{C}$ is given by the equation $y = \gamma(x)$ in a neighborhood of d . Therefore, we get a function $\tilde{g} : \bar{\mathbf{Q}} \rightarrow \mathbf{P}^1$ such that $\tilde{g}(x) = g(x, \gamma(x))$ for any x in a neighborhood of d_1 . Then, we can write in a neighborhood of d_1 , $\tilde{g}(x) = ((x - d_1)\tilde{f}_1(x), (x - d_1)\tilde{f}_2(x)) = (\tilde{f}_1(x), \tilde{f}_2(x))$ where $\tilde{f}_1$ and $\tilde{f}_2$ are functions which are non-zero at d_1 .

Conversely, if we are given a rational function s/t defined over $\mathcal{C}$ - where s and t are linear form - which has exactly simple zeros at a and b and simple poles at o and at a point e , then $e = a + b$. Indeed, suppose that we are given such a function s/t . Then, s and t have both three zeros (counted with multiplicity) over $\mathcal{C}$. Hence, the fact that s/t has two zeros and two poles implies that s and t share a common zero over $\mathcal{C}$. This zero is the intersection of $\mathcal{C}$, of (a, b) (by definition of s), and of (o, e) (by definition of t). Thus, $e = a + b$ comes from the definition of $+$.

Similarly, $x = (a + b) + c$ if and only if there exists a rational function h defined on $\mathcal{C}$ which has simple poles at a, b and c , double zero at o and simple zero at x . Such a function can be build as $(f_2 f_4)/(f_1 f_3)$ where f_3 is the linear form which is zero at $a + b$ and c , and f_4 the linear form which is zero at o and $(a + b) + c$.

Since the expression above is symmetric, we deduce $(a + b) + c = a + (b + c)$. $\square$

3.2 Explicit group law

Since the neutral element has been explicitly chosen as $(0, 1, 0)$ and the equation of $\mathcal{C}$ is written $y^2 = x^3 + \alpha x + \beta$ in the affine plane $z = 1$, we can give some explicit formula to calculate the sum of two rational points.

Proposition 3.2.1. *Let a and b be points of $\mathcal{C}$. Then $-a = (a_1, -a_2)$, and $a + b = (\lambda^2 - a_1 - b_1, -\lambda^3 + \lambda(a_1 + b_1) - \mu)$, where λ is equal to $\frac{a_2 - b_2}{a_1 - b_1}$ when $a \neq b$ and to $\frac{3a_1^2 + \alpha}{2a_2}$ when $a = b$, and μ is equal to $a_2 - \lambda a_1$.*

Proof. Let us consider a and b , two rational points.

Since $-a$ is the symmetric of P_1 with respect to the x -axis, $-a = (a_1, -a_2)$.

Let us write $y = \lambda x + \mu$ the equation of (a, b) (which is considered as the tangent of $\mathcal{C}$ in a when $a = b$). Then λ and μ are the same as in the proposition above.

Let us write $c = -(a + b)$, the third point of $\mathcal{C}(\mathbf{Q}) \cap (a, b)$.

According to the equations of $\mathcal{C}$ and (a, b) , we get the following equation:

$$0 = x^3 + \alpha x + \beta - (\lambda x + \mu)^2 = x^3 - \lambda^2 x^2 + (\alpha - 2\lambda\mu)x + (\beta - \mu^2).$$

This polynomial equation of degree 3 has a_1 , b_1 and c_1 as roots. Therefore, we get by relations between coefficients and roots:

$$a_1 + b_1 + c_1 = \lambda^2.$$

Moreover, $c_2 = \lambda c_1 + \mu$ by definition.

We deduce that $(a + b)_1 = c_1 = \lambda^2 - a_1 - b_1$ from which we deduce that $(a + b)_2 = -c_2 = -\lambda^3 + \lambda(a_1 + b_1) + \mu$. $\square$

From Theorem 3.2.1 we deduce by direct computations the following proposition:

Proposition 3.2.2. *Let a and b be distinct rational points. We have:*

$$(a + b)_1 + (a - b)_1 = \frac{2(a_1 + b_1)(\alpha + a_1 b_1) + 4\beta}{(a_1 - b_1)^2},$$

$$(a + b)_1(a - b)_1 = \frac{(a_1 b_1 - \alpha)^2 - 4\beta(a_1 + b_1)}{(a_1 - b_1)^2},$$

and

$$(2a)_1 = \frac{a_1^4 - 2\alpha a_1^2 - 8\beta a_1 + \alpha^2}{4(a_1^3 + \alpha a_1 + \beta)}.$$

4 Height theory

In this section, we introduce the height theory. The height of a point could be interpreted as its "algebraic complexity".

4.1 Weil height

Definition 4.1.1. Let $x \in \mathbf{P}^n(\mathbf{Q})$. Multiplying x by a scalar, it is always possible to suppose x written $x = (x_0, \dots, x_n)$ where x_i are integers prime together. Then, the height of x is $H(x) = \max_{i \in \llbracket 0, n \rrbracket} (x_i)$. We define also a logarithmic height h as $\ln(H)$.

Remark. The height of a point can be seen as its "complexity" from an algebraic point of view. For instance, consider $\frac{1}{2}$ and $\frac{501}{1000}$ (seen as the elements $(1, 2)$ and $(501, 1000)$ of $\mathbf{P}^1(\mathbf{Q})$). These are closed numbers but of really different heights (2 for the first one and 1000 for the second). Thus, the height of a rational x shows how high integers should be to describe x .

We give the following theorem, which is obvious when restricting ourselves to $\mathbf{Q}$ as we do (but be aware that height theory can be generalized to any number field):

Theorem 4.1.2 (Northcott, Kronecker). *The set $\{x \in \mathbf{P}^n(\mathbf{Q}), H(x) \leq X\}$ is finite for any real X .*

Now, let us state a second useful result:

Theorem 4.1.3. *The height h is symmetric and satisfies:*

$$\forall(a, b) \in \mathcal{C}(\mathbf{Q})^2, h(a + b) + h(a - b) = 2h(a) + 2h(b) + O(1).$$

Thus, h is almost quadratic. The next section aims at smoothing h to make it become quadratic.

4.2 Néron-Tate height

In this section, we try to give a definition of the height of a point canonical to $\mathcal{C}$. Actually, the height of a point could be defined via the ambient space $\mathbf{P}^n$. Alas, such a definition would not be intrinsic. Thus, this section aims at providing a height more suited to $\mathcal{C}(\mathbf{Q})$.

Lemma 4.2.1. *Let us pose S a set. Let $h : S \rightarrow \mathbf{R}$ and $f : S \rightarrow S$ be two maps satisfying $|h \circ f - dh| \leq C$ for a real d greater than 1 and a real C . Then, for all $x \in S$, the sequence $(d^{-n}h(f^n(x)))_{n \in \mathbf{N}}$ converges to a limit $\hat{h}_f(x)$ which satisfies:*

$$|h(x) - \hat{h}_f(x)| \leq \frac{C}{d-1},$$

and

$$\hat{h}_f(f(x)) = d\hat{h}_f(x).$$

Proof. Let $x \in S$. By the inequality hypothesis, we have at $f^{k-1}(x)$ for any positive integer k :

$$-\frac{C}{d^k} \leq \frac{1}{d^k}h(f^k(x)) - \frac{1}{d^{k-1}}h(f^{k-1}(x)) \leq \frac{C}{d^k}.$$

By a telescoping sum, we get for any natural integers n and m such that $n < m$:

$$-\frac{C}{d^n(d-1)} \leq \frac{1}{d^m}h(f^m(x)) - \frac{1}{d^n}h(f^n(x)) \leq \frac{C}{d^n(d-1)}.$$

Therefore, $(d^{-n}h(f^n(x)))_{n \in \mathbf{N}}$ is a Cauchy sequence and its limit $\hat{h}_f(x)$ exists. Now, letting m tend to infinity, we get (by setting $n = 0$):

$$-\frac{C}{d-1} \leq \hat{h}_f(x) - h(x) \leq \frac{C}{d-1}.$$

This is the first inequality we aimed to prove. For the second one, we have:

$$\hat{h}_f(f(x)) = \lim_{n \rightarrow \infty} \frac{1}{d^n}h(f^{n+1}(x)) = d \lim_{n \rightarrow \infty} \frac{1}{d^{n+1}}h(f^{n+1}(x)) = d\hat{h}_f(x).$$

□

We can now define a new height:

Definition 4.2.2. The canonical or Néron-Tate height of a rational point x is:

$$\hat{h}(x) = \lim_{n \rightarrow +\infty} \frac{h((2^n x)_1)}{4^n}.$$

Theorem 4.2.3 (Néron-Tate). *The height $\hat{h}$ satisfies $\hat{h}(x) = h(x) + O(1)$ for any rational point x and the parallelogram law.*

Remark. Before proving Theorem 4.2.3, observe that it implies that $\hat{h}$ is quadratic.

Proof. First of all, $\hat{h}$ is well defined thanks to Theorem 4.2.1 (take $d = 4$ and $f : x \mapsto 2x$). Furthermore, Theorem 4.2.1 gives immediately that $\hat{h}(x) = h(x) + O(1)$ for any rational point x . Let x and y be rational points. Then, we dispose of a constant C by Theorem 4.1.3 such that:

$$\forall n \in \mathbf{N}, -C \leq h(2^n(x+y)) + h(2^n(x-y)) - 2h(2^n x) - 2h(2^n y) \leq C.$$

We conclude by dividing by 4^n and letting n tend to infinity. $\square$

5 The Mordell-Weil Theorem

In this section, we draw our inspiration from Hindry [3]. Our goal is to prove the following theorem:

Theorem 5.0.1 (Mordell-Weil). *The group $(\mathcal{C}(\mathbf{Q}), +)$ is finitely generated.*

5.1 The Weak Mordell-Weil Theorem

The goal of this section is to prove a weaker form of the Mordell-Weil Theorem:

Theorem 5.1.1 (Weak Mordell-Weil). *The group $\mathcal{C}(\mathbf{Q})/2\mathcal{C}(\mathbf{Q})$ is finite.*

We assume that the equation of $\mathcal{C}$ can be written as follows:

$$y^2 = f(x) = (x - \alpha_1)(x - \alpha_2)(x - \alpha_3),$$

where α_1, α_2 and α_3 are rational. If this assumption does not hold, it is always possible to take $\mathbf{Q}(\alpha_1, \alpha_2, \alpha_3)$ instead of $\mathbf{Q}$ (here, we dealt only with the field $\mathbf{Q}$ but there exists a height theory for number fields also).

In addition, observe that roots of f are distinct since $\mathcal{C}$ is smooth. Indeed, the gradient of the curve at a point (x, y) is of the form $(-f'(x), 2y)$ and therefore, f' should not vanish at a non-zero rational.

We consider also $P_i = (\alpha_i, 0)$ for $i \in \{1, 2, 3\}$.

Definition 5.1.2. Let $\psi = (\psi_1, \psi_2, \psi_3) : \mathcal{C}(\mathbf{Q}) \rightarrow (\mathbf{Q}^*/(\mathbf{Q}^*)^2)^3$ be defined by:

$$\psi_i(x, y) = \begin{cases} 1 & \text{if } x = o, \\ (\alpha_i - \alpha_j)(\alpha_i - \alpha_k) & \text{if } x = P_i, \\ x - \alpha_i & \text{else.} \end{cases}$$

Remark. Since we consider elements modulo $(\mathbf{Q}^*)^2$ and $f(x)$ is a square for any element (x, y) of $\mathcal{C}(\mathbf{Q})$, we get:

$$(x - \alpha_1) = (x - \alpha_1)f(x) = (x - \alpha_2)(x - \alpha_3).$$

This justifies the values taken by ψ in the P_i .

Now we should prove three lemmas about ψ in order to conclude.

Lemma 5.1.3. *The map ψ is a homomorphism.*

Proof. Let us fix $i \in \{1, 2, 3\}$.

First of all, observe that for any rational points a, b and c , the equation $a + b + c = o$ is equivalent to saying that a, b and c are colinear. According to Theorem 3.2.1, $\psi_i(a) = \psi_i(-a) = \psi_i(a)^{-1}$ since $\mathbf{Q}^*/(\mathbf{Q}^*)^2$ is of exponent 2, for any rational point a . Then, it is sufficient to prove that $\psi_i(a)\psi_i(b)\psi_i(c) = 1$ for any rational points a, b and c such that $a + b + c = o$.

Thus, let us consider a, b and c , rational points such that $a + b + c = o$.

Let us consider D , the line of equation $y = \lambda x + \mu$ which contains a, b and c .

First, suppose $\{a, b, c\} \cap \{o, P_i\}$ empty. Then a_0, b_0 and c_0 are solutions of $f(x) - (\lambda x + \mu)^2 = 0$. Writing $x' = x - \alpha_i$, we get the equation $f(x' + \alpha_i) - (\lambda x' + \lambda \alpha_i + \mu)^2 = 0$. Hence, the polynomial $f(x' + \alpha_i) - (\lambda x' + \lambda \alpha_i + \mu)^2$ has $a_0 - \alpha_i, b_0 - \alpha_i$ and $c_0 - \alpha_i$ as roots. Since $f(\alpha_i) = 0$, the polynomial $f(x' + \alpha_i)$ has no constant term. Consequently, the constant term of the polynomial $f(x' + \alpha_i) - (\lambda x' + \lambda \alpha_i + \mu)^2$ is $-(\lambda \alpha_i + \mu)^2$. By the relations between coefficients and roots, we get:

$$(a_0 - \alpha_i)(b_0 - \alpha_i)(c_0 - \alpha_i) = (\lambda \alpha_i + \mu)^2.$$

From this we deduce $\psi_i(a)\psi_i(b)\psi_i(c) = 1$.

If $c = o$, then the relation is obvious as explained above.

Suppose now a, b and c different from o and one of them equal to P_i . Without loss of generality, let us suppose $a = P_i$ and b and c different from P_i . Then, according to Theorem 3.2.1, we get:

$$(a + b)_1 = \left(\frac{y_1}{b_1 - \alpha_i} \right)^2 - b_1 - \alpha_i = \frac{f(x_1)}{(b_1 - \alpha_i)^2} - b_1 - \alpha_i.$$

Then, we get:

$$\psi_i(a+b) = (b_1 - \alpha_i)^2 \psi_i(a+b) = (b_1 - \alpha_i)((b_1 - \alpha_j)(b_1 - \alpha_k) - (b_1 + 2\alpha_i)(b_1 - \alpha_i)).$$

Observe that the first factor is $\psi_i(b)$. For the second one, observe that $\alpha_i + \alpha_j + \alpha_k = 0$ because of relations between coefficients and roots of f . Therefore, we get:

$$(b_1 - \alpha_j)(b_1 - \alpha_k) - (b_1 + 2\alpha_i)(b_1 - \alpha_i) = \alpha_j \alpha_k + 2\alpha_i^2.$$

Similarly, we have:

$$\psi_i(a) = \alpha_i^2 - \alpha_i\alpha_j - \alpha_i\alpha_k + \alpha_j\alpha_k = \alpha_j\alpha_k + 2\alpha_i^2 - \alpha_i(\alpha_i + \alpha_j + \alpha_k) = \alpha_j\alpha_k + 2\alpha_i^2.$$

Thus, we have $\psi_i(a)\psi_i(b) = \psi_i(a+b)$.

□

Lemma 5.1.4. *The kernel of ψ is $2\mathcal{C}(\mathbf{Q})$.*

Proof. The inclusion $2\mathcal{C}(\mathbf{Q}) \subset \ker(\psi)$ is clear because $\mathbf{Q}^*/(\mathbf{Q}^*)^2$ is of exponent 2 and ψ is a homomorphism according to Theorem 5.1.3.

Now, let us consider an element (x, y) of $\ker(\psi)$ written k . Then there exists (z_1, z_2, z_3) , a triple of elements of $\mathbf{Q}^*$ such that:

$$\forall i \in \{1, 2, 3\}, x - \alpha_i = z_i^2.$$

Observe that this include the case $x = \alpha_i$ since 0 is a square.

Since the roots of f are supposed distinct, there exists a unique solution (u, v, w) to the Vandermonde system:

$$\begin{pmatrix} 1 & \alpha_1 & \alpha_1^2 \\ 1 & \alpha_2 & \alpha_2^2 \\ 1 & \alpha_3 & \alpha_3^2 \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} z_1 \\ z_2 \\ z_3 \end{pmatrix}.$$

Observe that we can express powers of α_i as linear combination of 1, α_i and α_i^2 thanks to f . Therefore, considering the equation $x - \alpha_i = (u + v\alpha_i + w\alpha_i^2)$, we get:

$$u^2 - 2vw\beta - x + \alpha_i(2uv - 2vw\alpha - w^2\beta + 1) + \alpha_i^2(v^2 + w\alpha + 2uw) = 0.$$

Therefore, we get a polynomial equation of degree 2 which has α_1, α_2 and α_3 as roots. Therefore, its coefficients should all be zeros. We deduce then the following system:

$$\begin{cases} u^2 - 2vw\beta - x = 0, \\ 2uv - 2vw\alpha - \beta w^2 + 1 = 0, \\ v^2 + 2uw - \alpha w^2 = 0. \end{cases}$$

Multiplying the second line by w , the third by v and subtracting them, we get $v^3 + vw^2\alpha + \beta w^3 - w = 0$. Besides, if $w = 0$ then the third line implies that $v = 0$, and then the second one implies $1 = 0$ which is false. Consequently, we can divide the equation above by w^3 and get:

$$\left(\frac{v}{w}\right)^3 + \alpha\left(\frac{v}{w}\right) + \beta = \left(\frac{1}{w}\right)^2.$$

Therefore, the point $(\frac{v}{w}, \frac{1}{w})$, written h , belongs to $\mathcal{C}(\mathbf{Q})$.

Moreover, $k = 2h$ or $k = 2(-h)$. Indeed, according to Theorem 3.2.2 we have:

$$(2h)_1 = \frac{h_1^4 - 2\alpha h_1^2 - 8\beta h_1 + \alpha^2}{4h_2^2} = \frac{v^4 - 2\alpha v^2 w^2 - 8\beta v w^3 + \alpha^2 w^4}{4w^2}.$$

Then, according to the third line of the system above, we get:

$$\begin{aligned}
(2h)_1 &= \frac{(\alpha w^2 - 2uw)^2}{4w^2} + \frac{1}{4}(-2\alpha v^2 - 8\beta vw + \alpha^2 w^2) \\
&= \frac{1}{4}\alpha^2 w^2 - u\alpha w + u^2 - \frac{1}{2}\alpha v^2 - 2\beta vw + \frac{1}{4}\alpha^2 w^2 \\
&= u^2 - 2\beta vw - \frac{\alpha}{2}(v^2 - \alpha w^2 + 2uw),
\end{aligned}$$

so, according to the first and the third lines of the system, $(2h)_1 = x = k_1$.

Therefore, $|(2h)_2| = |k_2|$ and we get the result. $\square$

Lemma 5.1.5. *The image $\psi(\mathcal{C}(\mathbf{Q}))$ is finite.*

Proof. Let us pose $\Delta = ((\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3)(\alpha_2 - \alpha_3))^2$ the discriminant of f which is non-zero as explained above. Let us consider a finite set S of prime numbers and pose the ring of S -algebraic integers as follows:

$$\mathcal{O}_S = \{x \in \mathbf{Q}, \forall p \notin S, v_p(x) \geq 0\}.$$

Besides, we consider the subset of S -units (this is the set of invertible elements of $\mathcal{O}_S$):

$$\mathcal{O}_S^* = \{x \in \mathbf{Q}, \forall p \notin S, v_p(x) = 0\}.$$

We can define an euclidean division with the absolute value. In particular, $\mathcal{O}_S$ is principal. Moreover, up to adding elements to $\mathcal{O}_S$, we can suppose that Δ belongs to $\mathcal{O}_S^*$. Therefore, the $\alpha_i - \alpha_j$ belong to $\mathcal{O}_S^*$.

Let us consider a rational point $a = (x, y)$. We write $x = \frac{u}{v}$ and $y = \frac{w}{t}$ where u, v, w and t belong to $\mathcal{O}_S$ and $\gcd(u, v) = \gcd(w, t) = 1$ in $\mathcal{O}_S$. Then, we can rewrite $y^2 = f(x)$ as follows:

$$w^2 v^3 = t^2 (u - \alpha_1 v)(u - \alpha_2 v)(u - \alpha_3 v).$$

Because of the conditions on the greater common divisors above, v^3 divides t^2 and reciprocally. Hence, up to modifying v and t by a unit, we can suppose $v^3 = t^2$. Since 2 is relatively prime to 3, we dispose of an S -algebraic integer s such that $v = s^2$ and $t = s^3$. Then we get:

$$a = \left(\frac{u}{s^2}, \frac{w}{s^3} \right),$$

and

$$w^2 = (u - \alpha_1 s^2)(u - \alpha_2 s^2)(u - \alpha_3 s^2).$$

Any common divisor of $u - \alpha_1 s^2$ and $u - \alpha_2 s^2$ divides their difference $(\alpha_1 - \alpha_2)s^2$ and also the linear combination $(\alpha_1 - \alpha_2)u$. Since s^2 and u are relatively prime together, we deduce that such a divisor should divide $\alpha_1 - \alpha_2$. Since $\alpha_1 - \alpha_2$ is a unit, we deduce that $u - \alpha_1 s^2$, $u - \alpha_2 s^2$ and $u - \alpha_3 s^2$ are relatively

prime together. Moreover, since their product is the square w^2 , they all are squares up to a unit.

Hence, we have:

$$x - \alpha_i = \frac{u - \alpha_i s^2}{s^2} = \gamma_i r_i^2,$$

where γ_i are S -units and r_i belong to $\mathbf{Q}$.

Therefore, it is enough to show that $\mathcal{O}_S^*/(\mathcal{O}_S^*)^2$ is finite. Let us then pose $\gamma \in \mathcal{O}_S^*$. We can write $S = \{p_1, \dots, p_n\}$. Then, we write $|\gamma| = p_1^{\varepsilon_1} \dots p_n^{\varepsilon_n}$ where ε_i can be chosen as elements of $\{0, 1\}$. Hence, $\mathcal{O}_S^*/(\mathcal{O}_S^*)^2$ is finite. This concludes the proof. $\square$

Finally, we can give the proof of *Theorem 5.1.1*:

Proof of Theorem 5.1.1. By Theorem 5.1.3, ψ is a homomorphism, of kernel $2\mathcal{C}(\mathbf{Q})$ by Theorem 5.1.4. Hence, $\mathcal{C}(\mathbf{Q})/2\mathcal{C}(\mathbf{Q})$ is isomorphic with the image of ψ , which is finite according to Theorem 5.1.5. Therefore, $\mathcal{C}(\mathbf{Q})/2\mathcal{C}(\mathbf{Q})$ is finite. $\square$

5.2 Proof of the Theorem

Now, we only need the following lemma to finish the proof:

Lemma 5.2.1 (Descent). *Let G be an abelian group and q be a quadratic form over G . Suppose that $\{x \in G, q(x) \leq X\}$ is finite for all $X \in \mathbf{R}$ and that $G/2G$ is finite. Then G is finitely generated by $\{x \in G, q(x) \leq C\}$ where $C = \max_{x \in S} q(x)$ and where S is a set of representations modulo $2G$.*

Proof. Let $x \in G$.

If $q(x)$ were negative, then there would be infinitely many elements where q would be negative by homogeneity. Then, let us write $|x|_q = \sqrt{q(x)}$. Hence $|\cdot|_q$ satisfies the triangular inequality and for any natural integer m , $|mx|_q = m|x|_q$.

Suppose $q(x) > C$, where C is defined as above (Theorem 5.2.1). Let us build a sequence $(x_n)_{n \in \mathbf{N}}$ as follows:

- $x_0 = x$,
- $x_n = y_{n+1} + 2x_{n+1}$ where $y_{n+1} \in S$ is a representation modulo $2G$ of x_n .

Then we have for any natural integer n such that $q(x_n) > C$:

$$|x_{n+1}|_q = \frac{|x_n - y_{n+1}|_q}{2} \leq \frac{|x_n|_q + |y_{n+1}|_q}{2} \leq \frac{|x_n|_q + \sqrt{C}}{2} < |x_n|_q.$$

We immediately deduce that $(|x_n|_q)_{n \in \mathbf{N}}$ is decreasing until $|x_n|_q \leq \sqrt{C}$. Indeed, since $\{y \in G, q(y) < q(x_0)\}$ is finite, there exists a natural integer m such that $|x_m|_q \leq \sqrt{C}$ and so the above inequality cannot be stated.

Therefore, $x_0 = y_1 + 2(y_2 + 2(y_3 + \dots(y_m + 2x_m)\dots))$ and so x is generated by elements from $\{y \in G, q(y) \leq C\}$. $\square$

Finally, we can prove Theorem 5.0.1:

Proof of Theorem 5.0.1. Thanks to Theorem 5.1.1, $\mathcal{C}(\mathbf{Q})/2\mathcal{C}(\mathbf{Q})$ is finite. Hence we can apply Theorem 5.2.1 to the group $\mathcal{C}(\mathbf{Q})$ endowed with the height $\hat{h}$ which is a quadratic form and satisfies the hypothesis of the lemma according to Theorem 4.2.3 and to Theorem 4.1.2. $\square$

5.3 The Mordell group

Now that we have finally proved the marvelous Theorem 5.0.1, we are eager to compute Mordell groups for some examples. However, even though we can compute a set of generators of $\mathcal{C}(\mathbf{Q})$ when we are provided with $\mathcal{C}(\mathbf{Q})/2\mathcal{C}(\mathbf{Q})$ (Theorem 5.2.1 tells us how to do so), we do not know any algorithm which is guaranteed to compute $\mathcal{C}(\mathbf{Q})/2\mathcal{C}(\mathbf{Q})$ in a finite amount of time.

However, we know that any finitely generated abelian group G is isomorphic with $\mathbf{Z}^r \times T$, where r is the rank of G and T is a finite abelian group (isomorphic with a product of $\mathbf{Z}/n\mathbf{Z}$) called the torsion subgroup of G . Fortunately, thanks to Theorem 5.0.1, we can guess the form of the Mordell group and compute it in some concrete cases.

Here is an example where the group $\mathcal{C}(\mathbf{Q})$ has been identified:

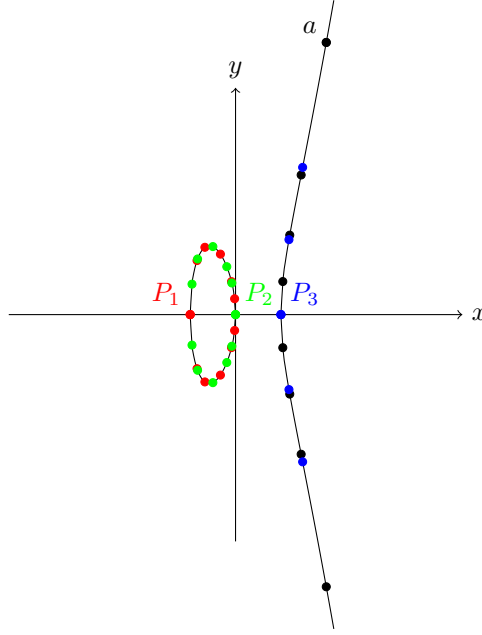


Figure 2: Mordell group of $y^2 = x^3 - 36x$

On Figure 2, $\mathcal{C}(\mathbf{Q})$ is generated by $P_1 = (-6, 0)$, $P_2 = (0, 0)$ and $P_3 = (6, 0)$ the points of order 2 and the point $a = (12, 36)$ of infinite order ($\langle a \rangle$ is black,

$\langle a \rangle + P_1$ is red, $\langle a \rangle + P_2$ is green and $\langle a \rangle + P_3$ is blue). Therefore, we get the isomorphism $\mathcal{C}(\mathbf{Q}) \simeq \mathbf{Z} \times \mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z}$. Hence, $\mathcal{C}(\mathbf{Q})$ is of rank 1 and its torsion subgroup is $\mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z}$.

Let us explain with a simpler example how the proof of Theorem 5.0.1 can help us to compute the Mordell group. We propose to compute the Mordell group of $y^2 = x^3 - x$. The first step is to compute $\mathcal{C}(\mathbf{Q})/2\mathcal{C}(\mathbf{Q})$. In order to do so, let $(\frac{u}{s^2}, \frac{w}{s^3})$ be a point of $\mathcal{C}(\mathbf{Q})$ as given by the proof of Theorem 5.1.5. This proof gives us $u = \gamma_0 r_0^2$, $u-1 = \gamma_- r_-^2$ and $u+1 = \gamma_+ r_+^2$ where γ_0 is prime with γ_- and γ_+ and $\gcd(\gamma_-, \gamma_+)$ is equal to 1 or 2. Since the product of γ_0 , γ_- and γ_+ is a square, $(\gamma_-, \gamma_0, \gamma_+)$ belong to $\{(1, 1, 1), (2, 1, 2), (-1, -1, 1), (-2, -1, 2)\}$. Hence, a set S of representations modulo $2\mathcal{C}(\mathbf{Q})$ is $\{o, (-1, 0), (0, 0), (1, 0)\}$. These points have a canonical height bounded by 8. Hence, we can check thanks to Theorem 5.2.1 that $\{(-1, 0), (0, 0), (1, 0)\}$ is a set of generators of $\mathcal{C}(\mathbf{Q})$. Therefore, $\mathcal{C}(\mathbf{Q})$ is of rank 0 and isomorphic with $\mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2\mathbf{Z}$.

Torsion subgroups of elliptic curves are well known as the following theorem shows:

Theorem 5.3.1 (Mazur). *The torsion subgroups of elliptic curves are exactly the fifteen following: $\mathbf{Z}/n\mathbf{Z}$ for $n \in \{1, \dots, 10\} \cup \{12\}$ and $\mathbf{Z}/2\mathbf{Z} \times \mathbf{Z}/2n\mathbf{Z}$ for $n \in \{1, 2, 3, 4\}$.*

On the opposite, there are still famous open problems concerning the rank of an elliptic curve. For example, Birch and Swinnerton-Dyer conjectured that the rank of an elliptic curve $\mathcal{C}(\mathbf{Q})$ is equal to the order of the zero of the Hasse-Weil function $L(\mathcal{C}, s)$ at $s = 1$. Observe that the rank of an elliptic curve can be zero as it is the case for $y^2 = x^3 + 1$ but can also be equal to 5 for a similar equation such as $y^2 = x^3 + 109858299531561$ ($\alpha = 0$ in both cases). Hence, the global form of the equation of an elliptic curve does not seem to give any information on its rank. Another interesting conjecture suggests that there could exist elliptic curves of arbitrarily high rank. The elliptic curve with the highest rank calculated hitherto is of rank 20 and given by the frightening following equation :

$$\begin{aligned} y^2 + xy + y = & x^3 - x^2 \\ & - 244537673336319601463803487168961769270757573821859853707x \\ & + 9617101820531830345462229792588068177... \\ & ...43270682028964434238957830989898438151121499931. \end{aligned}$$

References

- [1] Joseph H. Silverman. *The Arithmetic of Elliptic Curves*. Vol. 106. Graduate Texts in Mathematics. Springer-Verlag, 1986.
- [2] J. W. S. Cassels. *LMSST: 24 Lectures on Elliptic Curves*. London Mathematical Society Student Texts. Cambridge University Press, 1991.
- [3] Marc Hindry. *Arithmetics*. London: Springer, 2011.